

END TERM EXAMINATION

FIFTH SEMESTER [B.TECH] FEBRUARY 2023

Paper Code: ETCS303

Subject: Software Engineering

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions including Q.No.1 which is compulsory.

Assume missing data. Select one question from each unit.

- Q1 Attempt all questions: (2.5x10=25)
- a) Distinguish between failure and faults.
 - b) Compare functional & behavioral analysis models.
 - c) Write characteristics of software as a product.
 - d) Write short note on Spiral Model with its merits & demerits.
 - e) What are the different activities during software project planning?
 - f) Define Data flow diagram & Data Dictionary.
 - g) What are responsibilities and challenges of software engineers?
 - h) Identify in which phase of the software life cycle the following documents are delivered.
 - i) Architectural design
 - ii) Test plan
 - iii) Cost estimate
 - iv) Source code document
 - i) Differentiate between object oriented design and function oriented design.
 - j) List the principles of a software design.

UNIT-I

- Q2
- a) Which is more important-the product or process? Justify your answer (6.5)
 - b) Discuss Information flow Metrics. (6)
- Q3
- a) Explain Water fall Model. What are the problems that are sometimes encountered when the waterfall model is applied? (6)
 - b) Discuss the classification of Software Metrics. (6.5)

UNIT-II

- Q4
- a) Create ER diagram & DFD for Library Management Systems. Make assumptions & clearly state them. (6.5)
 - b) What are the risk management activities? Give top five risks in case of software development. (6)

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- Q5 a) Draw an ER and DFD diagram for university information System. (6.5)
b) Differentiate functional and non functional requirements and explain. (6)

UNIT-III

- Q6 a) Discuss the concept of cohesion and coupling. State the difference. (6.5)
b) How do you define Reliability? Discuss various models for reliability allocation. (6)
- Q7 a) Discuss Object Oriented Design in detail. (6.5)
b) Explain in detail about the characteristics and criteria for a good design. (6)

UNIT-IV

- Q8 a) What is software maintainability? How do you measure maintainability? (6.5)
b) Explain Reverse Engineering and Re-engineering. (6)
- Q9 a) What is the necessity of unit testing? Write down all unit test considerations. (6.5)
b) Write a note of (i) Black box testing (ii) Integration testing (iii) Decision table testing (6)
